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Earth and Planetary Science Letters 203 (2002) 755–767

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Pliensbachian magnetostratigraphy: new data from Paris Basin (France)

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Received 25 February 2002; received in revised form 6 July 2002; accepted 12 August 2002

Abstract

Sampling of an industrial drill string from the northeastern Paris Basin (Montcornet, France) provides early Jurassic magnetostratigraphic data coupled with biochronological control. About 375 paleomagnetic samples were obtained from a 145 m thick series of Pliensbachian rocks. A composite demagnetization thermal up to 300°C and an alternating field up to 80 mT were used to separate the magnetic components. A low unblocking temperature component (<250°C) with an inclination of about 64° is interpreted as a present-day field overprint. The characteristic remanent component with both normal and reversed antipodal directions was isolated between 5 and 50 mT. Twenty-nine polarity intervals were recognized. Correlation of these new results from the Paris Basin with data from the Breggia Gorge section (Ticino, southern Alps, Switzerland), which is generally considered as the reference section for Pliensbachian magnetostratigraphy, reveals almost identical patterns of magnetic polarity reversals. However, the correlation implies significant paleontological age discrepancies. Revised age assignments of biostratigraphic data of Breggia as well as an objective evaluation of the uncertainties on zonal boundaries in both Breggia and Moncornet resolve the initial discrepancies between magnetostratigraphic correlations and biostratigraphic ages. Hence, the sequence of magnetic reversals is significantly strengthened and the age calibration is notably improved for the Pliensbachian, a stage for which sections combining adequate magnetic signal and biostratigraphic constraints are still very few.
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Keywords: magnetostratigraphy; biostratigraphy; Lower Jurassic; correlation

1. Introduction

The magnetic reversal time scale is well known for post-late Jurassic times. Despite minor differences, a correlation between magnetostratigraphic polarity intervals and marine magnetic anomalies has been established on firm ground [1]. As the

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PII: S0012-821X(02)00898-1

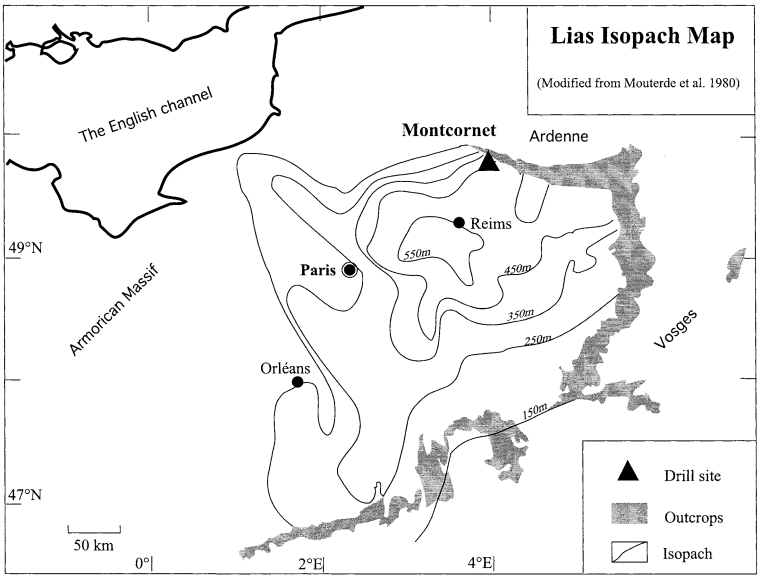


Fig. 1. Lias (Lower Jurassic) isopach map of the Paris Basin (after [12]).

record of oceanic magnetic lineations does not extend beyond late Jurassic time, magnetostratigraphy is the only method for investigating older changes in geomagnetic polarity. During the last two decades, several magnetostratigraphic data sets were obtained for the early–middle Jurassic [2–11]. Despite all of these efforts, there are still time intervals which need improved calibration between the sequence of polarity reversals and ammonoid zones within the early Jurassic. The increasing use of magnetostratigraphic correlation justifies the improvement of the calibration with ammonoid zones, and the new Pliensbachian data presented here may contribute to this endeavor. Although there are already data available for the Carixian (lower Pliensbachian) of the southern Alps [2], the sampling of an industrial drill string in the Paris Basin through rocks spanning the same time interval provides further constraints on the magnetic polarity time scale. From the bore core located on the northern side of the early Liassic transgressive embayment of the Paris Basin (Fig. 1) [12], previous magnetostratigraphic results were obtained for the Hettangian–Sinemurian interval, [11]. In this paper we present the magnetostratigraphic results obtained for the Pliensbachian (i.e., Carixian and Domerian) interval and compare them with the sequence from the Breggia Gorge in the southern Alps [2], which is so far considered to be the best documented sequence for this time interval.

2. Facies and sampling

There is a clear distinction between the lower calcareous and the upper shaly parts of the Liassic, which has long been recognized throughout most of the European Shelf and approximately coincides with the Carixian–Domerian boundary. In the Montcornet core, the transition occurs around 865 m (Fig. 2). The lower part of the section is essentially composed of bioclastic marls, with minor, interbedded bioclastic limestone, suggesting open platform depositional environments. The upper part is predominantly clayey, with, in stratigraphic succession, laminated mudstone, dark mudstone with calcareous and dolomitic nodules and a few intercalated bioclastic beds. This series ends around 786 m with a significant increase of clastic material (siltstone and sandstone with subordinate conglomerates) and erosional discontinuities. This clastic interval clearly stands out on the γ-ray log.

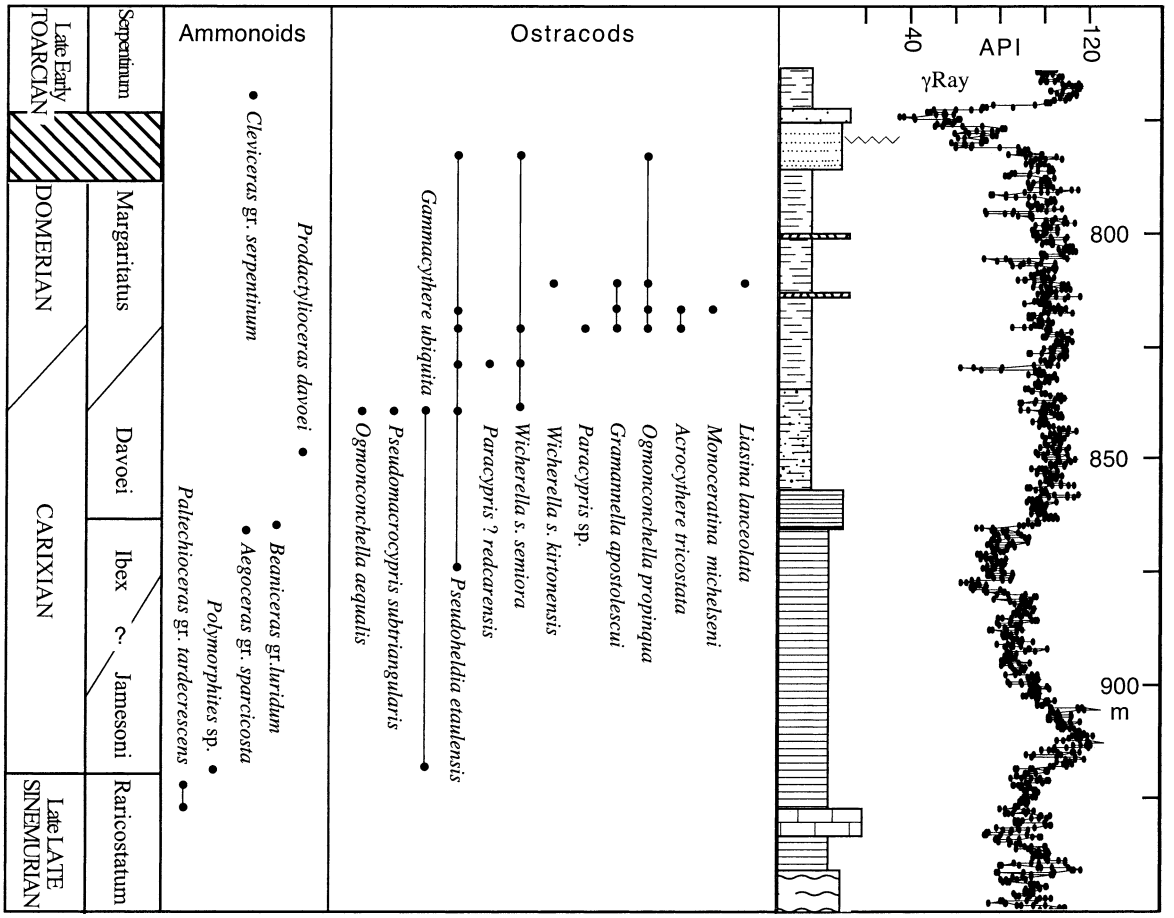


Fig. 2. Biostratigraphic age constraints (ammonoids and ostracods; see text for further explanations) and position of stage boundaries, lithology and γ-ray log (American Petroleum Institute) of the Montcornet core. Depth scale is in meters below surface. Shaded area indicates clastic input associated with important sedimentary discontinuities.

Core orientation is nearly vertical (280°/84°) as measured at 920 m depth using an inclinometer. Attitude of regional bedding is almost horizontal ($<180°/0.5°$), thus implying minimal bedding correction. Core segments 3 m long, 11 or 15 cm in diameter were recovered. An arbitrary upward directed orientation line was marked on each single or composite segment. Composite segments were restored when core fragments could be put back together without ambiguity about their internal rotation along breakages. Boundaries between successive core segments are not always preserved. Although composite segments without internal rotations may be as long as 9 m, shorter and mechanically less resistant claystone-dominated segments may not exceed a few decimeters in length. Along the 145 m of the core that corresponds to deposits of Pliensbachian age, the average sample spacing is about 30 cm, but due to varying quality of core recovery some gaps as long as 3 m sporadically occurred. Assuming a duration of 5.7 My for the Pliensbachian Stage [13,14] and a constant sedimentation rate, 1 m of rock amounts to 40 ky on average and, therefore, our sampling has an average resolution of 12 ky. Because of the high clay content, all paleomagnetic samples were drilled with compressed air as a cooling fluid.

3. Age control

Ammonoids of Sinemurian to Toarcian age were identified; however, none from the Domerian were recovered (Fig. 2). For this reason, ostracods were re-sampled in order to pin down the Carixian–Domerian boundary, which could not be placed with enough confidence on the basis of previously described ostracods and foraminifers (Andreieff, 1991, Bureau de Recherches Géologiques et Minières, unpublished report). The boundary between the late Sinemurian and the early Carixian is well constrained by the late Sinemurian *Paltechioceras* gr. *tardecrescens* at 920.7 m and by the early Carixian (Jamesoni Zone) *Polymorphites* sp. at 920.60 m. This is agreement with the FOD (first occurrence datum) of *Gammacythere ubiquita* found at 915 m. The boundary between the Jamesoni and Ibex is not well constrained, but the FOD of *Pseudoheldia etaulensis* (873 m) is usually considered a good marker of the base of the Ibex Zone [15]. This interval of uncertainty for the Jamesoni–Ibex boundary is indicated in Fig. 2 with a question mark. Further up, the Ibex–Davoei boundary is unambiguously placed on the basis of *Aegoceras* gr. *sparcicosta*, which typically occurs at this boundary.

The Carixian–Domerian boundary is defined on the basis of new ostracod samples. Typical Domerian ostracods such as *Ogmoconchella propinqua* and *Gramannella apostolescui* have their FOD at 820.48 m, whereas Carixian taxa such as *Gammacythere ubiquita* and *Pseudomacrocypris subtriangularis* have their LOD (last occurrence datum) at 838.3 m. Hence, the boundary can only be placed between these two levels as indicated in Fig. 2.

The LODs of typical Domerian ostracods occur at 781 m, within the clastic interval where neither younger late Domerian nor early Toarcian faunas have been documented. The next overlying ammonoid occurrence (*Cleviceras* gr. *serpentinum*) above the clastic interval, at 769.05 m, is diagnostic of the Serpentinum Zone of late early Toarcian. The clastic interval may thus encompass most of the late Domerian and the earliest Torcian.

4. Magnetic properties

In order to identify the magnetic carriers in these samples we applied the method described by Lowrie [16]. For representative selection of

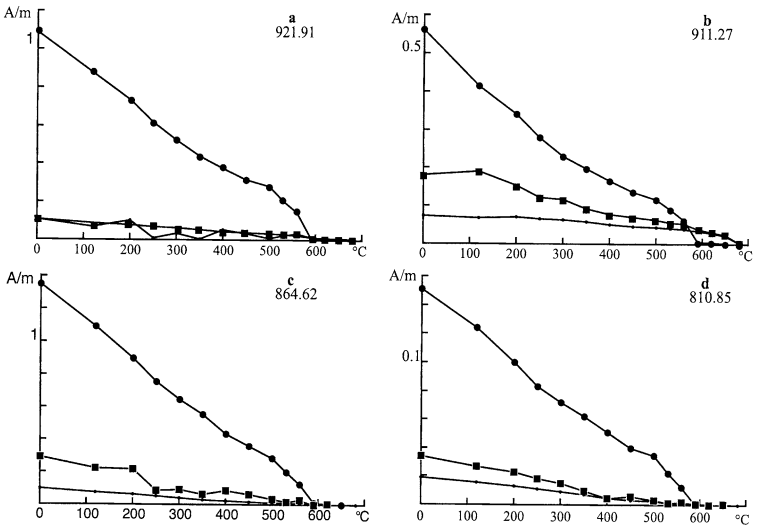


Fig. 3. Three-component IRM thermal demagnetization [16]. Solid circles indicate the soft component, 0.12 T field; squares indicate the medium component, 0.4 T field; small circles indicate the hard component, 1.2 T field. In most specimens, the soft component shows unblocking temperatures around 580°C, the Curie temperature of magnetite. Coexisting with magnetite is a high coercive component with unblocking temperatures of 680°C, most likely hematite.

samples a composite IRM (isotherm remanent magnetization) was subsequently thermally demagnetized (Fig. 3). Magnetic fields of 1.2, 0.4 and 0.12 T were successively applied to perpendicular axes prior.

The high coercivity component accounts for 5% of the remanence. It is completely unblocked at temperatures of 700°C, suggesting the presence of hematite (Fig. 3a,b) rather than goethite, which demagnetized below 120°C. The medium blocking field component accounts for 5% and 20% of the remanence. Unblocking temperatures of 700°C (Fig. 3b) indicate low coercivity hematite, but the intermediate unblocking temperature of 250–400°C (Fig. 3c and d) may indicate iron sulfides [16] or coarse-grained magnetite [17]. The low blocking field component is dominant and usually corresponds to a maximum unblocking temperature of about 580°C, which indicates that magnetite is the dominant carrier. The intensity of this soft component decreases progressively between 25°C and 500°C and then abruptly between 500°C and 580°C (Fig. 3a–d). The <500°C unblocking temperatures indicate multidomain magnetite, but the higher unblocking temperatures show that there is a significant proportion of single domain magnetite. Therefore, the behavior of the soft component, as presently observed on the demagnetization diagrams, may result from a mixture of different grain sizes of the magnetic carriers. It is well known that single domain grains of magnetite better record magnetic remanence than multidomain grains. Hence, the remanent magnetization of the largest grains tends to be viscous, and if only coarse grains are to occur in the rock, the overprint by the present magnetic field will be the only component. Taking into account a thermal gradient of 3°C/100 m in the Paris Basin [18] and a present maximal burial depth of 900 m, any viscous remanence acquired during the last few million years under a maximal temperature of 50°C should be removed when heating in the laboratory to 300°C (e.g., [19]). As a consequence, only samples having a well

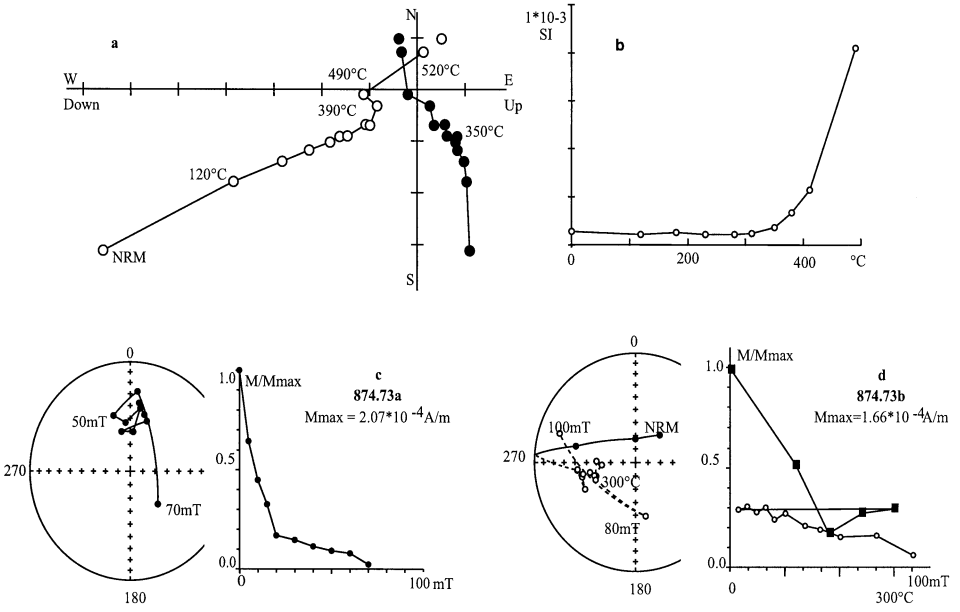


Fig. 4. Examples of typical demagnetization diagrams. (a) Orthogonal vector projection. Horizontal coordinates are arbitrary and vertical coordinates are with respect to the drill core axis. (b) Low field susceptibility during heating. (c,d) Comparative demagnetization for two twin specimens, equal-area projection and intensity of magnetization during demagnetization. Solid/open symbols refer to lower/upper hemispheres respectively. (c) AF (alternating field, mT is milliTesla) technique; (d) combined thermal and AF technique.

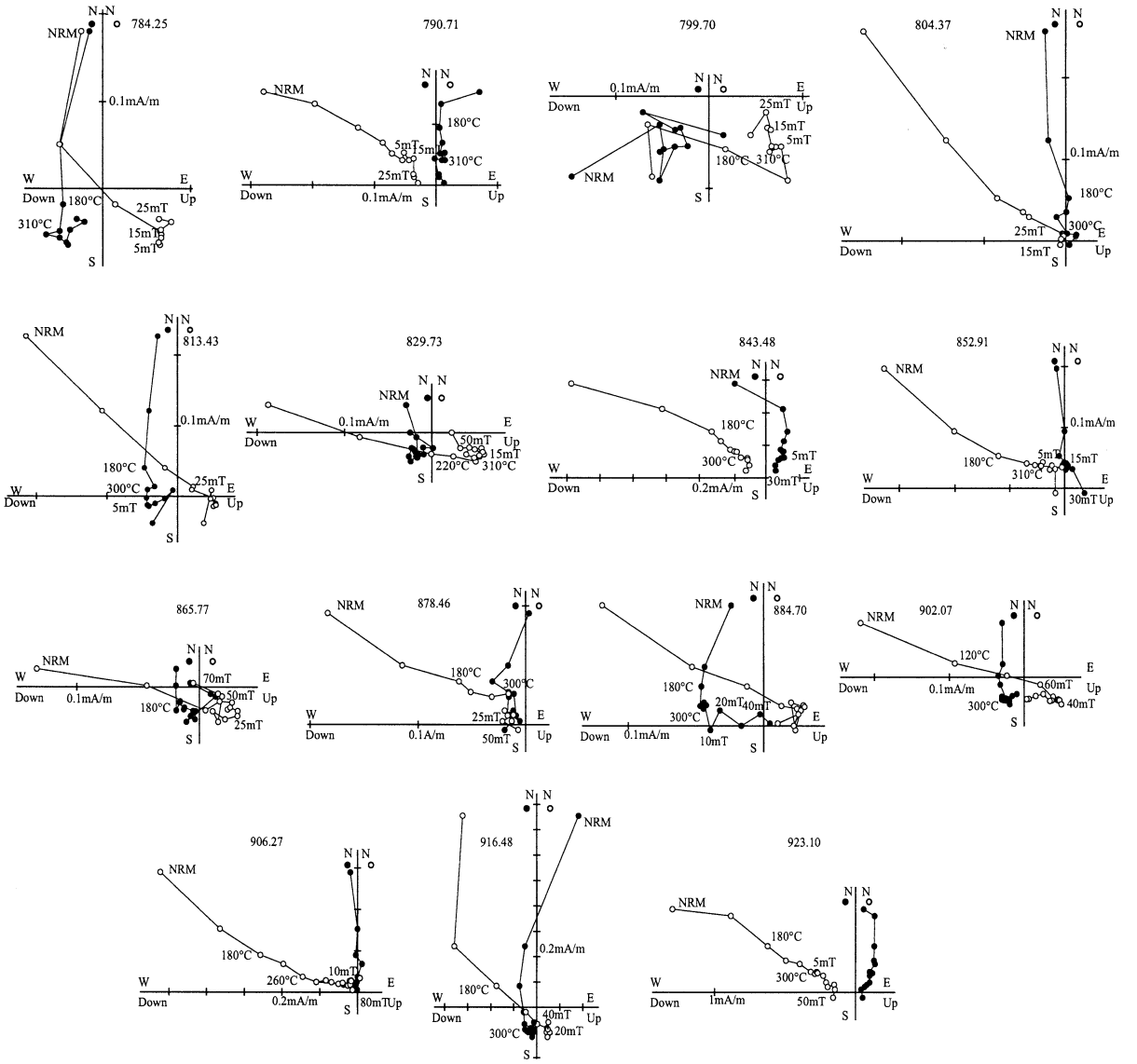


Fig. 5. Orthogonal vector projection of the combined thermal and AF demagnetizations for representative specimens. The declination is restored using the present field overprint.

defined magnetization persisting after heating to 300°C are worth being further demagnetized in order to isolate the characteristic magnetization.

5. Analysis of natural remanent magnetization (NRM)

The remanent magnetization of the samples was measured using a CTF cryogenic magnetometer in the shielded paleomagnetic laboratory at the Institut de Physique du Globe de Paris. NRM intensity ranges from 0.1 to 1 mA/m.

With thermal demagnetization 20–40% of NRM intensity in random direction is removed at 120°C. A well defined (linear) lower unblocking temperature component is removed between 120°C and 300–350°C (Fig. 4a). This component

is always normal in polarity and decreases linearly without drop. This observation disqualifies primary pyrrotite as carrier for this component and suggests coarse-grained magnetite instead. Magnetization becomes generally unstable at higher temperatures. A rapid increase of the bulk magnetic susceptibility above 350°C (Fig. 4b) indicates alteration of the magnetic mineralogy. As pyrite is observed in the samples, such changes can be explained by their oxidation. In one sample (Fig. 4c) demagnetization in alternative fields (AF) yielded only a single component. Following the procedure described by Yang et al. [11], its twin sample (Fig. 4d) was treated by hybrid demagnetization, i.e. a five-step heating to 300°C followed by progressive AF demagnetization up to 30–60 mT. This hybrid procedure yielded two distinct components and was therefore applied for routine treatment of all samples.

A stable component is defined for 70% of the specimens between 5 and 35–50 mT following thermal demagnetization to 300°C (Fig. 5). Directions were calculated by least square regression [20]; usually lines were constrained to pass origin. For 20% of the specimens, directions could not be calculated by means of least square regression because demagnetization paths from these samples seem to fit a great circle, which suggests a change from normal to reverse polarity during demagnetization [21]. For the remaining 10% of the specimens, demagnetization yielded blurred paths which cannot be objectively interpreted. Most of these specimens are located between 895 and 900 m deep. For this 5 m section of the core, no stable component was defined.

6. Interpretation of magnetic data

The random direction component removed by 120°C was probably acquired during drilling, storage and specimen sampling. The lower temperature component removed between 120 and 300°C is always normal in polarity. Application of the method of Enkin and Watson [22] to this component yields a mean inclination of $63.2 \pm 1.3°$. This value is in relatively good agreement with that of the expected present-day dipolar field at the drilling site ($D = 0$, $I = 67°$). This lower temperature component is interpreted to be a viscous overprint, acquired during the last 8×10^5 years (Brunhes Chron) and carried by coarse-grained magnetite.

The lower temperature component was then

Table 1
'Site' mean of stable component directions

Depth (m)	n	Dec	Inc	K	α_{95}
770–790	16	25.8	62.9	6.4	15.9
	20	213.9	−54.2	7.5	12.8
790–810	6	14.1	50.5	21.1	14.9
	26	213.7	−43.7	5.3	13.5
810–830	10	10.3	45.0	10.0	16.1
	22	216.2	−51.6	5.2	15.0
830–850	32	30.6	51.7	14.7	6.9
850–870	42	16.7	54.0	9.9	7.4
	13	235.7	−51.3	9.6	14.1
870–890	17	22.2	63.6	3.0	24.9
	52	211.3	−63.3	4.5	10.3
890–910	11	355.5	53.3	8.9	16.2
	41	218.7	−61.9	6.4	9.5
910–925	38	36.5	58.8	11.0	7.3
	6	226.4	−59.8	17.2	17.7
Mean of normal directions	8	18.2	55.6	69.5	6.7
Mean of reversed directions	7	219.5	−55.4	90.4	6.4
Total mean	15	28.1	55.9	56.3	5.1

n, number of samples; Dec, declination; Inc, inclination; K, precision parameter; α_{95}, radius of the circle of 95% confidence.

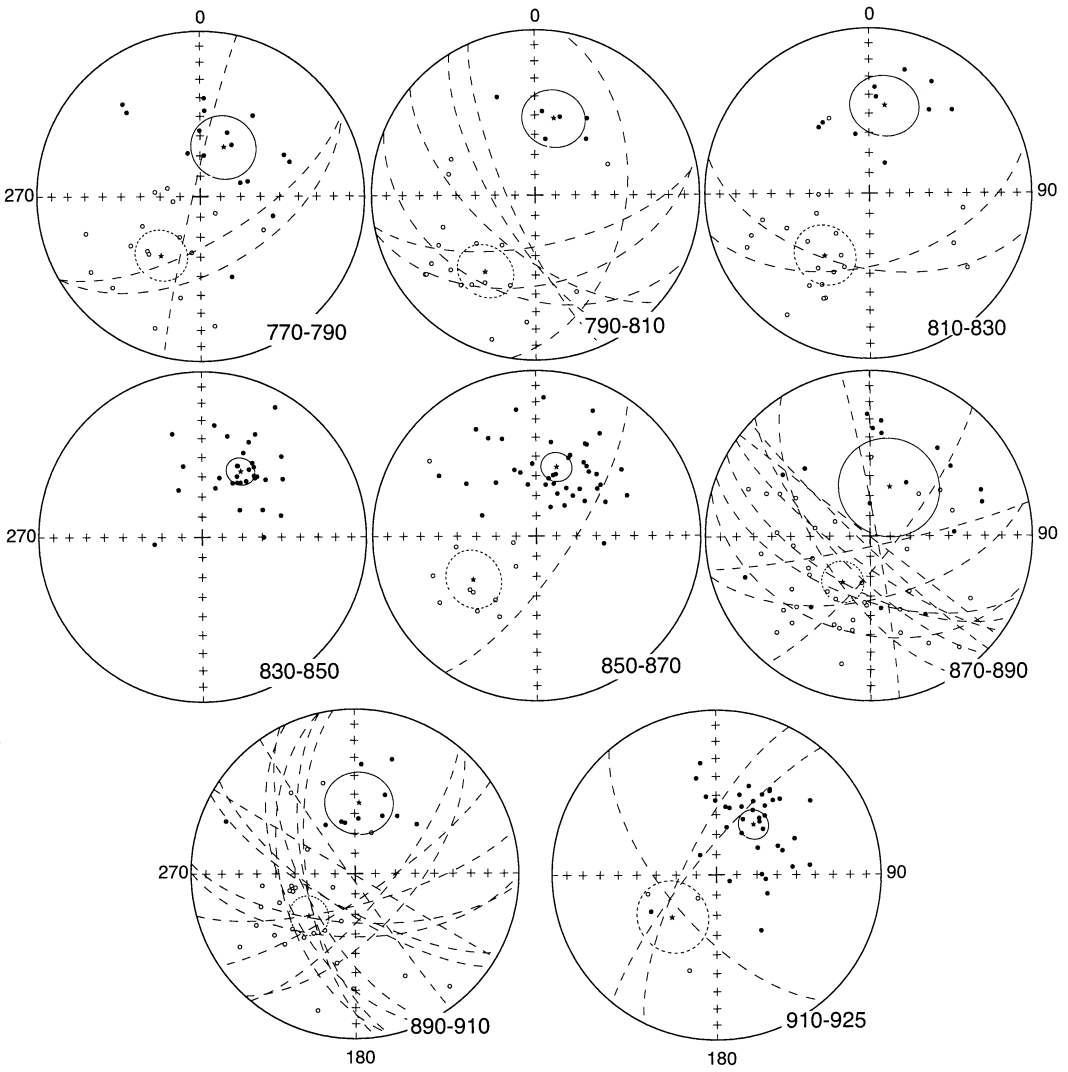


Fig. 6. Equal-area projection of characteristic components, demagnetization circles and means for each composite core segment (Table 1). Solid (open) symbols refer to the down (up) directions and only the upper halves of the demagnetization circles are shown.

used to restore the initial direction of the primary component. At 920 m depth, the drill string dips 84° toward 280° (W). Under this coordinate system, with the drill string being non-vertical, the present-day dipole field direction is $D = 14°$ and $I = 67°$. Therefore, to place the individual sample directions in this reference system, the low temperature and primary components must be rotated until the low temperature component coincides with an azimuth of 14°. Once the data are put in the proper reference frame, a correction for the dip of the drill string (6° with a strike of 190°) must be made to restore a geographic reference frame.

Eight groups of segments or 'sites', 20 m long each, were defined arbitrarily. For each 'site', we plotted all of the reoriented stable components or demagnetization circles (Fig. 6) and calculated the mean directions (Table 1). For all 'sites', in particular between 870 and 910 m, the directions of

characteristic components and the orientations of demagnetization circles are somewhat scattered, but no streaking trend can be defined. The scatter may have been caused by a hidden high stability component of random direction, probably acquired during drilling. Directions from higher demagnetization steps that seem to fit a great circle are interpreted as having been magnetized in reversed polarity and subsequently overprinted by an inseparable normal polarity acquired in the present magnetic field or perhaps in Jurassic time.

Only one 'site' uniformly shows stable components of normal polarity, in the seven other 'sites' both normal and reverse polarities exist. Mean directions of normal and reverse polarity components are not always strictly antipodal. These distributions may be largely influenced by the poor separation between the characteristic magnetic components and the present-day and drilling overprint. However, it is worthy of note that there are almost as many samples with normal polarity (171) as samples with reversed polarity (175), which suggests that biases in normal directions may compensate for biases in reversed directions in the mean combining both polarities.

Fig. 7 shows mean directions calculated for the normal and the reverse characteristic components of each 'site' as well as their corresponding global means of $D = 18.2°$, $I = 55.6°$, $\alpha_{95} = 6.7°$ for normal directions and $D = 219.5°$, $I = -55.4°$, $\alpha_{95} = 6.4°$ for reversed directions. The overall mean combining both polarities yields the following: $D = 28.2°$, $I = 55.9°$, and $\alpha_{95} = 5.1°$. Regional bedding attitude is almost horizontal (dip $< 0.5°$), thus requiring no bedding correction. The corresponding virtual geomagnetic paleopole is at 115.2°E, 66.0°N, $A_{95} = 6.6°$.

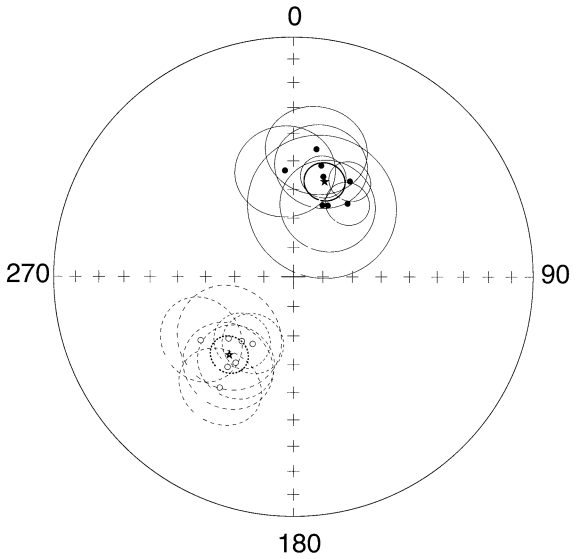


Fig. 7. Equal-area projection of mean characteristic remanent magnetization directions from each composite core segment, using the same convention as in Fig. 6.

7. Magnetostratigraphic results

Declination and inclination, weak field susceptibility, and γ-ray values are plotted as a function of the core depth in Fig. 8. Variations in weak field susceptibility and γ-ray logs reflect the changing properties of the lithological succession. The absence of correlation between the variations of these parameters suggests that the stable magnetic components are of primary origin, except around 865 m. At this particular point, there is a simultaneous change in magnetic polarity, an increase of approximately 100% of the weak field susceptibility and a marked increase in γ-ray values. This change occurs within 20 cm of very poorly consolidated rock, but no discontinuity is observed in the original core. Whatever the significance of this abrupt change, and without being able to exclude definitively the coincidence of a change in sedimentary regime with a change of polarity, the presence of a depositional hiatus remains possible.

Paleomagnetic directions of the samples were converted into virtual geomagnetic poles (VGP). VGP as well as biostratigraphic constraints are plotted in Fig. 9 as a function of the core depth. The specimens for which remagnetization circles were used are also shown (as x's) on this log. It is worth noting that most of these samples are found within or at the boundaries of reversed polarity zones. This clearly supports the interpretation that these samples have been magnetized in reversed polarity and subsequently contaminated by a normal polarity. However, a demagnetization circle was nevertheless fit to specimen 913.89. As this sample is stratigraphically brack-

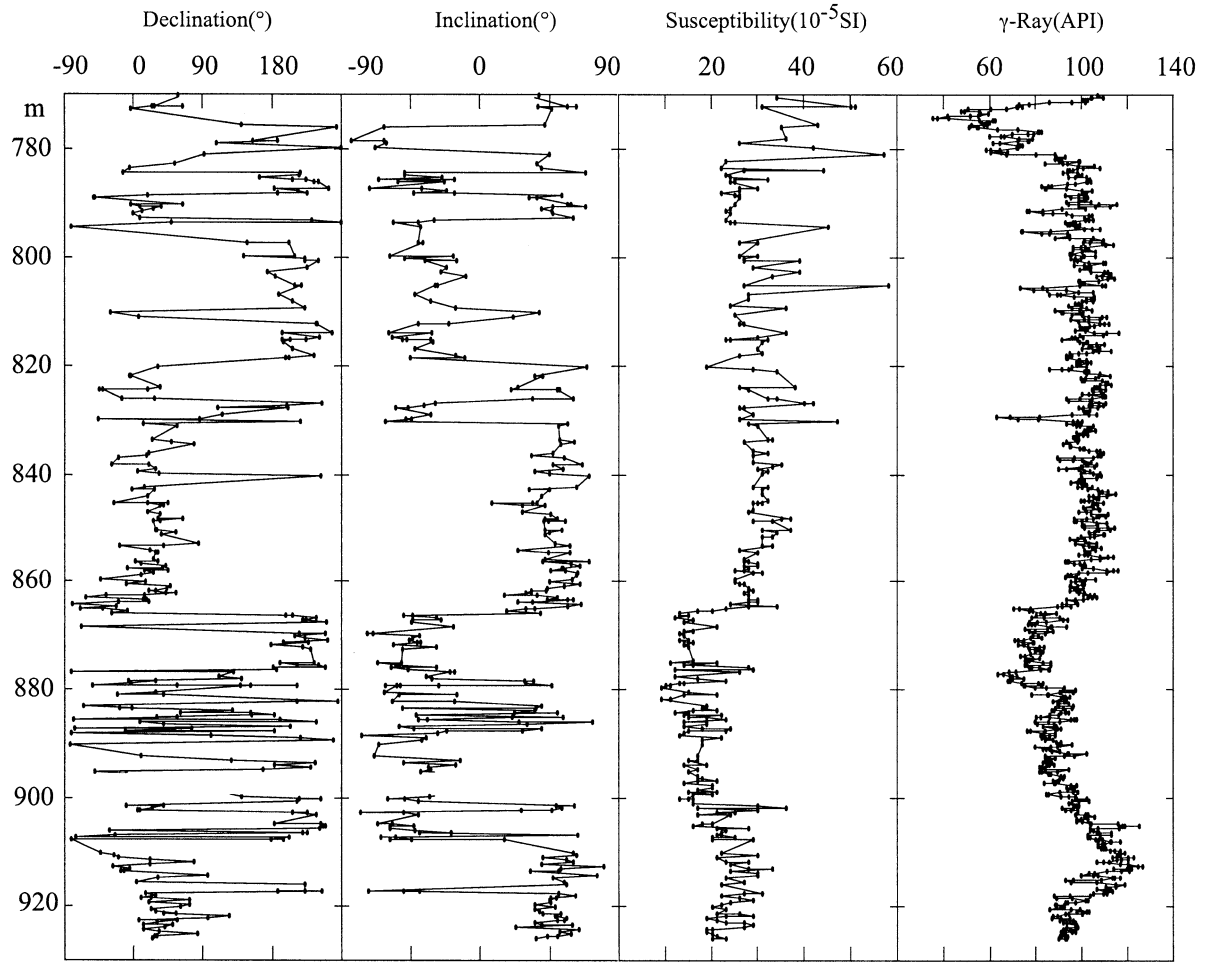


Fig. 8. Declination, inclination, weak field susceptibility and γ-ray versus depth of the Pliensbachian segment of the Montcornet core.

eted between samples showing relatively low VGP latitude values, it may suggest that a polarity reversal occurred at 913.89 m. In all other cases, the recognition of magnetic polarity intervals having characteristic directions was obtained from at least two samples. Twenty-nine polarity intervals are recognized.

8. Correlation with Breggia Gorge section and conclusion

The Breggia Gorge section (Ticino, southern Alps, Switzerland) is the best documented section for Pliensbachian magnetostratigraphy [2]. However, the comparison with our new results from the Paris Basin reveals both similarities and contradictions. With the exception of the late Domerian, whose sedimentary record is incomplete in Montcornet, an almost identical pattern of magnetic polarity reversals emerges between the two sections, despite the fact that their stratigraphic thicknesses differ by a factor 2 (see Fig. 10). The two sections share a sequence as follows: near the base, a predominantly normal polarity interval (A) with a few reversed chrons in its middle part, a second interval (B) dominated by reversed polarity with several intercalated short normal

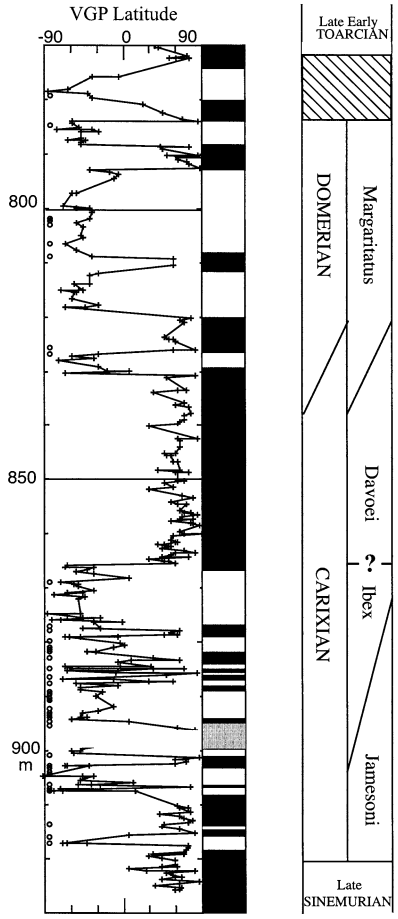


Fig. 9. Depth variations of VGP latitude (open circles indicate samples for which demagnetization circles were applied), magnetic polarity (normal: black; inverse: white; no data: gray) and ammonite zonation of the Pliensbachian from the Moncornet drill core (Paris Basin). Unidentified ammonoid zones and uncertain ages are shown. Depth scale is in meters below surface.

chrons, a third interval (C) almost entirely of normal polarity with the exception of a few short reversed chrons in its uppermost part, and a fourth interval (D) essentially of reversed polarity with two normal chrons.

Minor dissimilarities in the pattern of reversal sequences between the two sections can be readily explained by the uneven quality of the sedimentary record (sedimentary rates, gaps, etc.) and the varying quality of core recovery. For instance, a gap of about 3 m long (from 820.76 to 823.72 m) in the Montcornet core may explain why the C interval includes two short reversed chrons at Breggia and only one at Montcornet.

Although the two magnetic reversal sequences agree generally well, there are significant biostratigraphic age discrepancies, for which further explanation is needed. The main age difference lies with interval C. Its age is reported to be late Ibex Zone at Breggia but Davoei Zone at Montcornet. It also ensues that interval D is exclusively of Margaritatus Zone age at Moncornet, whereas it is of Davoei Zone and early Margaritatus Zone age in Breggia. This led us to re-evaluate the age control at Breggia, whose critical ammonoid data were provided by Wiedenmeyer ([23], see p. 39 and plate 33). These critical ammonoids are included in Fig. 10. In Breggia, *Uptonia jamesoni* is the only reported ammonoid indicative of the Jamesoni Zone. However, all other ammonoids found stratigraphically below its occurrence are actually younger. There are two possible explanations: either faulting or miss-identification. The critical specimen of *U. jamesoni* illustrated by Wiedenmeyer ([23], plate 2, figures 2 and 3) is badly deformed and could also be alternatively assigned to *Aegoceras*, a genus of Ibex and Davoei age. *Crucilobiceras gemmelaroi* is here assigned to *Metaderoceras*, whose age is within the Ibex Zone. *Protogrammoceras carixiense* also indicates an age not older than Ibex Zone. Therefore, none of the ammonoids from Breggia reported by Wiedenmeyer [23] can be assigned unambiguously to the earliest Carixian Jamesoni Zone, nor to the latest Carixian Davoei Zone. Then, we further argue that the boundary between the Davoei and Margaritatus Zones, as drawn by Wiedenmeyer [23] and subsequently adopted by Horner and Heller [2], cannot be placed with accuracy on the basis of the ammonoid record. *Fuciniceras lavinianum* unambiguously indicates a Margaritatus age. Hence, the Carixian–Domerian boundary must be significantly lowered down in the Breggia section. In combination with these revised age assignments, the objective evaluation of uncertainties about the position of zonal boundaries in both sections eliminates the initial discrepancies between magnetostratigraphic correlations and biostratigraphic ages for the early Pliensbachian

(Carixian). Ignoring intervals of uncertainty which accompany most zone boundaries can often lead to erroneous dating and correlations.

Hence, based on the new sequence of magnetic reversals and biostratigraphic data obtained from Moncornet and the revised biostratigraphic data of Breggia, an improved, composite magnetostratigraphy can be proposed for the entire Pliensbachian stage (Fig. 11). Moncornet provides the early Pliensbachian (Carixian) part of the composite magnetostratigraphic scale (intervals A, B, C) while Breggia provides the late Pliensbachian (Domerian) part of the scale (intervals D, E). Uncertainties about the exact positions of the Jamesoni–Ibex Zone boundary and the Carixian–Domerian boundary are clearly shown in the figure and they must be taken into account when using magnetostratigraphy as a correlation tool for the Pliensbachian. These uncertainties directly reflect the present amount and quality of data and could be easily narrowed down if new data were to become available.

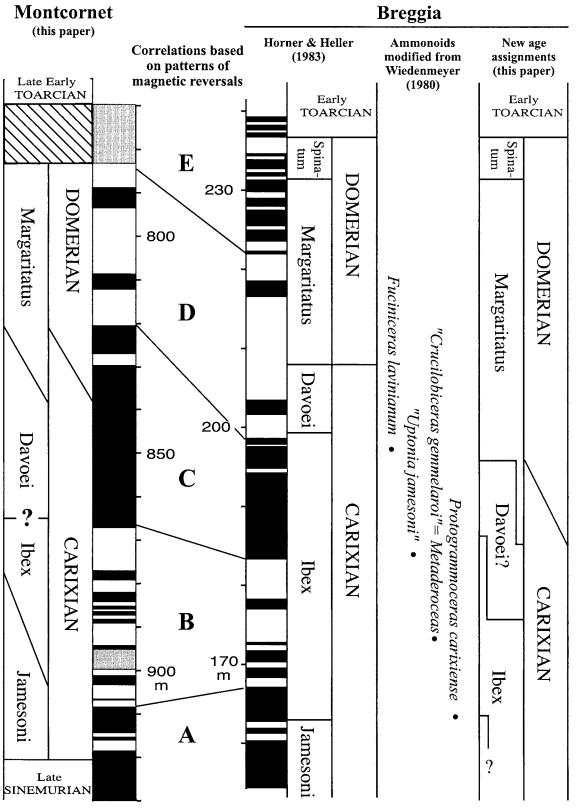


Fig. 10. Correlations between Montcornet and Breggia Gorge sections. From left to right: magnetostratigraphy and age control of the Montcornet section (this paper), correlations based on patterns of magnetic reversals, magnetostratigraphy and age assignments of the Breggia Gorge section [2], ammonoid data from Breggia (modified from Wiedenmeyer [23]), and revised age assignments for Breggia.

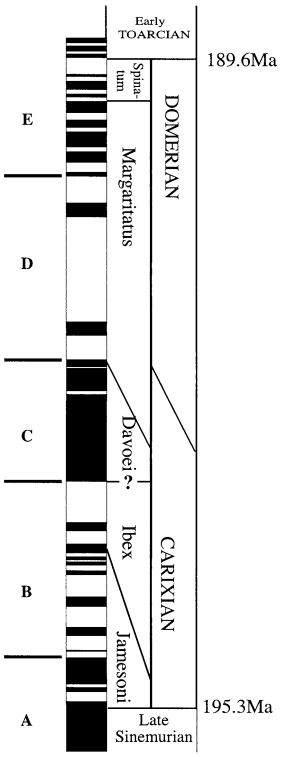


Fig. 11. Composite magnetostratigraphic scale for the Pliensbachian.

Acknowledgements

This research was supported by the Laboratory of Paleomagnetism at the Institut de Physique du Globe, Paris. Core samples and unpublished reports (micropaleontology, γ-ray profile) were generously made available by ANDRA. Core samples are archived at the ANDRA. Stuart Gilder and Randolph Enkin contributed usefully to the English language. M.L. Osete, G. Muttoni and an anonymous reviewer formally reviewed the manuscript. IPGP contribution 1831.*[AC]*

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